

Diseño de un controlador para un sistema de segundo orden.

$$V_e(t) = R i_1(t) + 2 \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_2(t) - i_2(t)]$$

$$2 \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de la place.

$$V_e(s) = R I_1(s) + 2s[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)]$$

$$2s[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico.

$$V_e(s) = 2s + 2R I_1(s) - (2s + R) I_2(s)$$

$$I_1(s) = \frac{1}{2s + R} \left(2s + \frac{1}{Cs} + 3R \right) I_2(s) = I_2(s) \left(\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} \right)$$

$$V_s(s) = \left(R + \frac{1}{Cs} \right) I_2(s) = \frac{CRs + 1}{Cs} I_2(s)$$

$$V_e(s) = \left[(2s + 2R) \frac{(CLs^2 + 3CRs + 1)}{CLs^2 + CRs} - 2s + R \right] I_2(s) - (2s + R) I_2(s)$$

Funcion de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs} I_2(s)}{\frac{3CLR s^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)} = \frac{(CRs + 1)(Ls + R)}{3CLR s^2 + (5CR^2 + L)s + 2R}$$

$$\frac{V_o(s)}{V_e(s)} = \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R}$$

- * Error en estado estacionario
- * Estabilidad lazo abierto
- * Bosquejo de respuesta (transitorio - estado estacionario)
- * Modelo de ecuación integro diferencial

$$\lim_{s \rightarrow 0} 1 - \frac{(LRs^2) + (CR^2 + L)s + R}{3LRs^2 + (5CR^2 + 2L)s + 2R} = 1 - \frac{R}{2R} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$R = 100 \text{ K}\Omega, L = 390 \text{ mH}, C = 470 \text{ nF}$$

$$a = 3CLR = 0.05499$$

$$b = (5CR^2 + L) = 23,500,000$$

$$c = 2R = 200 \text{ K}$$

$$\begin{cases} \lambda_1(t) = -0.008610 \\ \lambda_2(t) = -427.350427 \end{cases}$$

Sobreamortiguada



transitorio - estacionario

Ecuación - Integro diferencial

El modelo se expresa despejando las variables dependientes, es decir las corrientes

$$V_e(t) = \left[2Ri_1(t) + 2 \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$2 \frac{d[i_1(t) + (-i_2(t))]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + \frac{1}{C} \int \frac{I_{sc}}{dt}$$

$$I_2(t) = \left[L \frac{d(i_2(t) - i_1(t))}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - 2 \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = \left[2 \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$